SE3020 – Distributed Systems
BSc (Hons) in Information Technology Specialized in Software Engineering
Year 3 – Semester 1
2026 – Assignment 1

Assignment Duration – 5 Weeks

Title: Building an AI-Enabled Smart Healthcare Appointment & Telemedicine Platform using Microservices

Note: This assignment carries a weightage of **25%** towards the final grade. It is a **group project** with **3 to 4 members**.

You have been asked to develop a **cloud-native healthcare platform** similar to real-world telemedicine systems such as **Channeling.lk, oDoc,** or **mHealth**, which allows patients to book doctor appointments, attend video consultations, upload medical reports, and receive AI-based preliminary health suggestions. Below are the requirements provided by the client and/or the Business Analyst.

Requirements:

- **Web/Mobile Interface:** Develop a **web/mobile interface** where patients can browse available doctors, book appointments, and attend video consultations. Ensure the interface is **user-friendly and supports various devices**.

- **Patient Management Service:** Implement a service where patients can register, manage their profiles, upload medical reports/documents, and view their medical history and past prescriptions.
    - ***Patient role*** – Register, manage profile, book appointments, upload reports, view prescriptions, attend video consultations.

- ***Admin role*** **–** Manage user accounts, verify doctor registrations, and oversee platform operations and financial transactions.

- **Doctor Management Service:** Develop a service where doctors can manage their profiles, set availability schedules, conduct video consultations, and issue digital prescriptions.
  - ***Doctor role*** – Manage profile and availability, accept/reject appointment requests, conduct telemedicine sessions, issue digital prescriptions, and view patient-uploaded reports.

- **Appointment Service:** Implement a service that allows patients to search for doctors by specialty, book appointments, modify or cancel bookings, and track appointment status in real time.

- **Telemedicine Service (Video Session Integration):** Implement a secure video consultation module that enables patients and doctors to connect in real time. Integrate a **third-party video/conferencing API** such as **Agora, Twilio, or Jitsi Meet**.

- **Payment Service:** Integrate **secure online payment gateways** for consultation fees. Use **Sri Lankan third-party services** such as **PayHere, Dialog Genie, or FriMi** (or internationally recognized services such as **Stripe or PayPal (sandbox environment)**).

- **Notification Service:** Upon successful appointment booking or consultation completion, patients and doctors should receive **confirmation via SMS and email**. Utilize **third-party SMS and email services** for sending notifications.

- **AI Symptom Checker Service (Optional Enhancement):** Develop an optional AI-powered service where patients can input their symptoms and receive **preliminary health suggestions** and recommended doctor specialties. Integrate a suitable **AI/ML API or model** for this purpose.

You may add new functionalities other than those mentioned in the description.

Implementation:

1. **Based on the provided requirements, develop a set of RESTful web services** to implement the healthcare platform. You may choose any technology stack to implement the services. Ensure the services are designed following **REST principles**, maintaining **scalability, security, and performance**.
2. You must use the **Microservices architecture** to develop/integrate the API. Ensure that you use **Docker and Kubernetes**. If you are using any other tool for **Microservices orchestration/integration**, you may justify that in the report and during the viva.
3. Develop an **asynchronous web client** using any **JavaScript framework** that supports asynchronous programming (such as **Angular, React, etc.**) or use **regular jQuery + AJAX**. However, for the scope of this assignment, implementing just an **asynchronous web client** is sufficient.
4. Use appropriate **security/authentication mechanisms** to uniquely identify each user and authenticate them. There should be **three roles:**
    - **Patient** – Browse doctors, book appointments, attend video consultations, upload medical reports, and receive prescriptions.
    - **Doctor** – Manage availability, conduct consultations, and issue digital prescriptions, view patient records.
    - **Admin** – Manage user accounts, verify doctor registrations, and handle platform operations.

Deliverables:

1. A text file called **submission.txt**, containing a **GitHub repository link** with all the source code. The source code should contain the **backend services, client, and any other relevant source/resource files** (e.g., database scripts), arranged in a proper directory structure.
2. The **submission.txt** should contain a **YouTube video link** of a presentation/demo of the project. Each member may use **a maximum of 3 minutes** to explain their contribution, so the total video length **should not exceed 12 minutes**.
3. A **readme.txt** document, listing down the steps to deploy the above deliverables.
4. A **members.txt** file, containing the names, registration numbers, and the IDs of the group members.
5. A **report in PDF format (report.pdf)**. The report should include:
    - A **high-level architectural diagram** showing the services and their interconnectivity.
    - A list of **interfaces (NOT the user interfaces, but the service interfaces)** exposed by each service.
    - A brief explanation of **each workflow** used in the system (you may use design diagrams of your choice to do this).
    - **Details about authentication/security mechanisms** adopted.
    - **Individual contributions** of each group member.

You may use **code snippets** in the report to explain the above.

The **report must have an appendix** with all the code that you have written (**excluding the auto-generated code**). **Do not paste screenshots of the code in the appendix**; instead, copy the **code as text**. If screenshots are added in the appendix, **only the minimum mark may be offered**.

Note: All reports will be uploaded to **Turnitin** for plagiarism checking. If the **Turnitin similarity** is **above 20%**, marks will be penalized.

Submission Details:

- **All files should be uploaded in a single ZIP archive**. The ZIP file name should be **Group ID_DS-Assignment.zip**. **Only one member** needs to upload the submission.
- **Submission Deadline: 11th week of the semester**.